

Guruvishnu Nallamadugu

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SUMMARY:

Mechanical Design & Manufacturing Engineer with 3+ years of experience in mechanical design, manufacturing support, process improvement, and equipment development. Experienced in CAD design, GD&T, DFM/DFA, 3D printing, prototyping, BOMs, ECO/ECN, and root cause analysis. Proven ability to improve manufacturing efficiency, develop practical engineering solutions, and support products from design and prototyping through manufacturing, testing, and validation.

TECHNICAL SKILLS:

CAD & Mechanical Design: Creo, Siemens NX, CATIA V5, SolidWorks, GD&T, Tolerance Stack-Up, DFM/DFA.

Prototyping & Testing: 3D Printing, Prototype Development, Tooling & Fixture Design, Testing & Validation.

Quality & Problem Solving: Root Cause Analysis (RCA), 5-Why, DFMEA, Troubleshooting.

Production & Configuration: BOM Management, ECO/ECN, Production Support, Manufacturing Documentation.

Programming & Analysis: MATLAB, Python.

Manufacturing: Lean Manufacturing, Six Sigma, Process Optimization, SOP Development, Assembly Process Improvement.

EXPERIENCE:

Design Engineer | Mississippi State University | Starkville, MS

Jan 2026 - Present

- Design and develop custom research equipment and mechanical assemblies for laboratory and field applications, translating researcher's functional requirements into practical engineering solutions using 3D CAD modeling and mechanical design.
- Redesigned and modified existing research equipment to reduce overall footprint, optimize space utilization, and simplify field installation, including development of a compact 6-in-1 instrumentation assembly integrating multiple environmental and soil measurement devices into a single system.
- Apply Design for Manufacturing (DFM) and Design for Assembly (DFA) principles to component design, considering material selection, dimensions, tolerances, fastening methods, manufacturability, accessibility, and ease of assembly.
- Develop functional prototypes using 3D printing to evaluate fit, form, component integration, and functionality; iteratively refine CAD designs based on prototype testing, dimensional constraints, and researcher feedback.
- Conduct field testing and design validation alongside research teams by comparing measurements from prototype integrated instruments with reference equipment to verify measurement consistency and ensure the mechanical design does not interfere with sensor accuracy or performance.

Graduate Teaching Assistant | Mississippi State University | Starkville, MS

Jan 2024 – Dec 2025

- Served as a Graduate Teaching Assistant for Mechanics of Materials for 4 semesters, leading weekly recitation sessions and mentoring 100+ students each semester.
- Assisted students with laboratory experiments, connecting theoretical mechanics concepts with practical engineering applications and real-world problems.
- Supported exam preparation, proctoring, and grading, while providing guidance on problem-solving and course concepts.

CAD Designer | Tata Sikorsky | India

Dec 2021 – Dec 2022

- Designed flight control cable pulleys and mechanical guide brackets using Siemens NX and SolidWorks, supporting aerospace manufacturing and assembly requirements.
- Improved manufacturing cycle time by 8% by optimizing designs for automated production cells.
- Performed tolerance stack-up analysis (GD&T) to ensure manufacturability and eliminate assembly defects.
- Designed and evaluated tooling and fixtures to improve assembly efficiency and support production readiness.
- Collaborated with manufacturing and supplier teams and applied DFMEA/DFSS methods to improve product quality and reduce design and production risks.

Manufacturing Engineer | Bharat Electronics Limited (BEL) | India

Dec 2020 – Nov 2021

- Supported manufacturing and assembly of cold-storage chambers and HVAC equipment, adapting component layouts to chamber design and space constraints.
- Developed SOPs and conducted time studies to identify production bottlenecks, contributing to a 25% reduction in manufacturing cycle time.
- Managed and interpreted BOMs and supported first-article validation, ensuring accurate product configuration and manufacturing documentation.
- Implemented ECO/ECN design changes, updating BOMs, drawings, and manufacturing processes to support production readiness and configuration control.
- Performed RCA, 5-Why analysis, and corrective actions to resolve manufacturing issues, reduce defects, and improve equipment reliability.

PROJECTS:

Celestial Mechanics-Based Asteroid Orbit Optimization for Future Space Resource Utilization.

Research Project

May 2024 – Present

- Developed a computational framework in Python to model and visualize two- dimensional asteroid trajectories using principles of celestial mechanics.
- Retrieved real orbital elements and ephemeris data from the NASA Jet Propulsion Laboratory Small-Body Database for near-Earth asteroids.
- Simulated asteroid orbit propagation to identify optimal intervention windows and determine the minimum ΔV required to redirect candidate asteroids into stable Earth, Moon, or Mars vicinity orbits.
- Performed engineering simulations in ANSYS to evaluate force application strategies, orbital stability, and mission feasibility.
- Applied numerical methods and optimization techniques to minimize propulsion requirements while maintaining long-term orbital stability.
- Produced 2-D visualizations of orbital transfers to compare trajectory alternatives and support mission planning for future in-space resource utilization.

EDUCATION:

•Master of Science in Aerospace Engineering | Mississippi State University | **Dec 2025**

•Bachelor of Science in Aerospace Engineering | Chandigarh University | **June 2022**